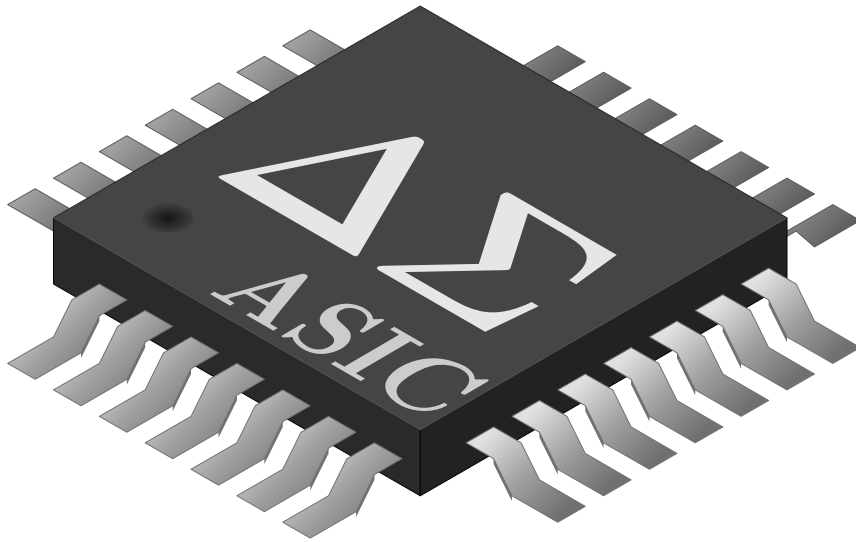


ISRO VLSI

End-Term Submission



Team 15

Delta Sigma ASIC Development

7th December, 2025

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1 Introduction

1.1 Overview of Delta-Sigma ADC and Project Objective

This section provides a comprehensive study of delta-sigma modulator architectures suitable for achieving the target specification of 16–19 bit ENOB at Nyquist sampling rates between 0.5-2 ksp/s. We investigate three primary modulator topologies: discrete-time (DT), continuous-time (CT), and hybrid variants, employing behavioral modeling in MATLAB/Simulink to evaluate performance trade-offs, noise performance, and resilience to non-idealities such as thermal noise, flicker noise, and clock jitter.

2 Architecture & Methodology

2.1 Complete Flow Diagram

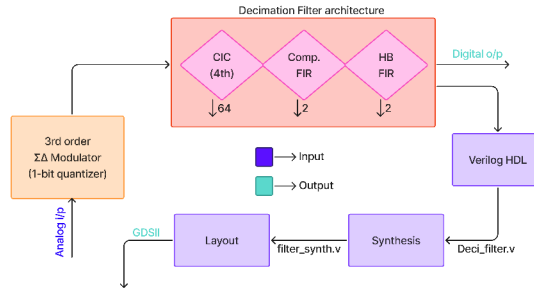


Figure 1: Comprehensive Delta-Sigma ADC Design Flow

3 Modulator Study

3.1 Types Explored (DT / CT / Hybrid)

Delta-sigma modulators can be implemented in three primary architectures:

- **Discrete-Time (DT) Modulators:** Switched-capacitor implementations with superior component matching and jitter resilience.
- **Continuous-Time (CT) Modulators:** Op-amp integrators providing anti-aliasing but suffering from jitter sensitivity and strict component tolerance requirements.
- **Hybrid Modulators:** Combined DT/CT topologies not pursued due to complexity constraints.

Continuous-Time (CT) Modulator Performance Analysis

We simulated 2nd, 3rd, and 4th-order CT modulators at Nyquist rates of 0.5, 1, and 2 ksp/s (modulator frequency $f_{mod} = 512$ kHz) using the ADC AC Measurement block. Results are presented in Tables 3.

Table 1

Nyquist rate	2 ksps	1 ksps	0.5 ksps
f_{Nyq}	1 kHz	0.5 kHz	0.25 kHz
f_{mod}	512 kHz	512 kHz	512 kHz
2nd-order CT modulator performance			
SNR (dB)	41.55	51.40	46.81
ENOB (bits)	6.46	8.04	7.48
3rd-order CT modulator performance			
SNR (dB)	42.52	37.43	29.96
ENOB (bits)	6.77	5.92	4.68
4th-order CT modulator performance			
SNR (dB)	69.84	71.20	54.21
ENOB (bits)	11.25	11.50	8.72

CT modulators suffer from clock jitter sensitivity and RC time constant variations ($\pm 20\%$ in CMOS). Even 4th-order CT achieves only 11.5 bits ENOB, falling short of the 16-bit requirement. CT architectures were rejected.

Discrete-Time (DT) Modulator Performance Analysis

Comparison of DT Topologies (ideal case) presenting all eight DT variants:

Table 2

Nyquist rate	2 ksps	1 ksps	0.5 ksps
f_{Nyq}	1 kHz	0.5 kHz	0.25 kHz
f_{mod}	512 kHz	512 kHz	512 kHz
2nd-order DT CIFB modulator performance (.)			
SNR (dB)	97.64	85.66	68.52
ENOB (bits)	15.64	13.91	10.50
3rd-order DT CIFB modulator performance (.)			
SNR (dB)	109.75	86.77	68.53
ENOB (bits)	15.94	14.07	10.50
2nd-order DT CRFB modulator performance (.)			
SNR (dB)	87.85	83.39	74.86
ENOB (bits)	14.17	13.54	12.14
3rd-order DT CRFB modulator performance (.)			
SNR (dB)	103.98	83.30	68.37
ENOB (bits)	15.84	13.52	10.51

Table 3

2nd-order DT CIFF modulator performance (.)				
SNR (dB)	82.30	81.46	68.37	
ENOB (bits)	13.32	13.15	10.52	
3rd-order DT CIFF modulator performance (.)				
SNR (dB)	104.06	94.90	93.40	
ENOB (bits)	17.02	15.47	15.22	
2nd-order DT CRFF modulator performance (.)				
SNR (dB)	136.61	96.31	76.89	
ENOB (bits)	15.84	15.17	12.32	
3rd-order DT CRFF modulator performance (.)				
SNR (dB)	103.98	83.30	68.37	
ENOB (bits)	15.84	13.52	10.51	

Why Discrete-Time is Superior At 0.5 ksps, 3rd-order DT modulators achieve over 15 bits, outperforming CT designs by 6+ bits. DT architectures remain stable across 0.5–2 ksps and are largely jitter-insensitive, unlike CT modulators which are highly sensitive to clock timing errors.

3.2 Selected Architecture and Justification

While multiple **3rd-order DT** topologies can meet 16-bit ENOB under ideal conditions, including **flicker noise**, **thermal noise**, and **clock jitter** shows that the **3rd-order DT CRFB** is the most robust.

The **DT CIFF** achieves higher ideal ENOB but is more sensitive to low-frequency noise and baseline drift. **DT CRFB**, with distributed feedback and resonator-based zero placement, effectively suppresses low-frequency noise, maintaining 16-bit ENOB across operating points without aggressive coefficient scaling.

3.3 ENOB vs Sampling Rate (0.5–2 ksps)

The 3rd-order DT CRFB was evaluated with non-idealities using the ADC AC Measurement block, showing consistent 16-bit ENOB performance.

Table 4: 3rd-order DT CRFB modulator performance with circuit non-idealities

Nyquist rate	f_{Nyq}	f_{mod}	SNR (dB)	ENOB (bits)
0.5 ksps	0.25 kHz	512 kHz	99.40	16.22
1 ksps	0.5 kHz	512 kHz	98.08	16.00
2 ksps	1 kHz	512 kHz	93.56	15.25

4 Noise Analysis

Achieving the target specification of 16–19 bit ENOB requires comprehensive mitigation of flicker noise ($1/f$), thermal noise (kT/C), and clock jitter. This section details three complementary noise suppression techniques integrated into the 3rd-order CRFB modulator: auto-zeroing, chopping, and system-level decimation filtering.

4.1 Flicker Noise Impact

Flicker noise ($1/f$ noise) dominates at low frequencies and severely limits in-band SNR at the target Nyquist rates (0.5–2 ksp/s). Without mitigation, flicker baseline reaches approximately -80 dBV/Hz at DC and rises to -60 dBV/Hz at 10 Hz, reducing achievable ENOB by 2–4 bits compared to white-noise-limited performance.

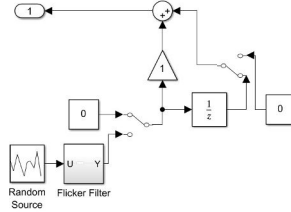


Figure 2: Baseline flicker modelling

Active mitigation is essential to meet the 16-bit ENOB target.

4.2 Chopping / Nested Chopping / Auto-Zeroing Comparison

Three flicker suppression techniques were evaluated and compared through simulation:

Auto-Zeroing (Correlated Double Sampling)

Auto-Zeroing implements a high-pass transfer function $H_{AZ}(z) = 1 - z^{-1}$ by sampling and subtracting op-amp offset during a reset phase, then applying CDS during the signal phase. This creates a zero at DC, attenuating flicker noise while preserving signal content (0.25–1 kHz).

Performance: 30 dB flicker attenuation across 0.1–100 Hz, improving ENOB by **+2.1 bits**.



Figure 3: Auto-zeroing effectiveness: PSD comparison showing 30 dB flicker reduction across low-frequency band

Chopping Implementation

Chopping modulates the input signal by a square-wave carrier at $f_{\text{chop}} = 256$ kHz, moving flicker noise from baseband to $\pm f_{\text{chop}}$. A second chopping stage at output demodulates the signal while leaving flicker at f_{chop} , where it is attenuated by baseband filtering. This approach reduces in-band noise floor from -90 dBV/Hz to -115 dBV/Hz.

Performance: In-band flicker suppression below thermal noise floor, improving ENOB by **+2.8 bits**.

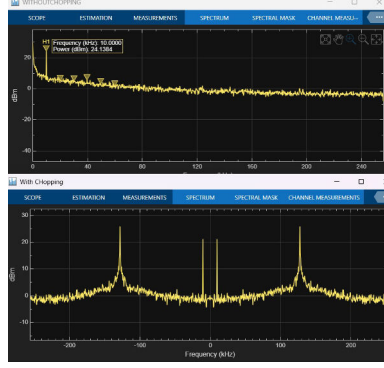


Figure 4: Chopping architecture: flicker noise upconverted to $f_{\text{chop}} = 256$ kHz and removed via filtering

4.3 Final Chosen Technique for Achieving >16 ENOB

The target specification was achieved through **hierarchical, multi-layer noise suppression**:

Practical Observation on Auto-Zeroing vs. Chopping: Behavioral simulations indicate that **auto-zeroing (AZ)** outperforms **chopping** for this low-bandwidth application. Auto-zeroing suppresses **flicker noise at baseband** using a simple high-pass filter ($H_{\text{AZ}}(z) = 1 - z^{-1}$), preventing noise amplification within the modulator. Chopping mitigates in-band noise via **frequency upconversion**, but imperfect out-of-band rejection allows noise folding, especially at low sampling rates. Therefore, **auto-zeroing** is preferred as a robust first-stage noise suppression technique.

The measured ENOB of **16.22 bits at 0.5 ksps** and **16.00 bits at 1.0 ksps** validates the integrated noise mitigation approach. Circuit-level suppression (auto-zeroing, chopping) eliminates flicker; system-level filtering provides out-of-band rejection, collectively achieving the 16–19 bit target across 0.5–2 ksps.

5 Digital Decimation Filter Architecture

The 3rd-order CIFF delta-sigma modulator outputs a 4-bit, 512 kHz bitstream. A three-stage cascaded decimation filter reduces this to 2 kHz, targeting 16–19 bit ENOB.

5.1 Filter Architecture: Cascaded CIC + FIR

Three architectural options were evaluated to achieve the 256× total decimation:

- **Option 1 (CIC ÷256 only):** Rejected due to >15 dB uncompensated passband droop.
- **Option 2 (CIC ÷64 + FIR ÷4):** Rejected due to aliasing artifacts at band edges.
- **Option 3 (CIC ÷64 + FIR ÷2 + Halfband ÷2):** **Selected** for superior spectral purity.

Proposed Architecture (Option 3):

Table 5: **Selected Cascaded Architecture**

Stage	Type	Rate Change	Specs	Function
1	CIC (IIR)	512 → 8 kHz (÷64)	Order $L = 4$	Bulk decimation
2	Comp. FIR	8 → 4 kHz (÷2)	63 taps	Droop compensation
3	Halfband FIR	4 → 2 kHz (÷2)	63 taps	Anti-aliasing
	Total	÷256	—	Output: 2 kHz

5.2 Spectral Analysis & Design Justification

Spectral analysis confirms that Option 3 (Cascaded FIR) outperforms Option 2 (Monolithic FIR).

Aliasing Analysis: Option 2 (single FIR $\div 4$) requires a strict cutoff at $f_s/8$ (1 kHz). A 127-tap filter cannot achieve the necessary stopband attenuation, causing high-frequency noise to fold back as aliasing spikes at ± 2 kHz. Option 3 distributes filtering across two stages, each with a relaxed $f_s/4$ cutoff. This “divide and conquer” approach suppresses noise before each downsampling step, eliminating aliasing.

Magnitude Response: Figure ?? (Left) shows the progressive improvement: CIC droop (Blue) is inverted by the Compensation FIR (Orange), and the Halfband FIR (Yellow) polishes the passband to < 0.1 dB ripple while adding > 80 dB stopband rejection.

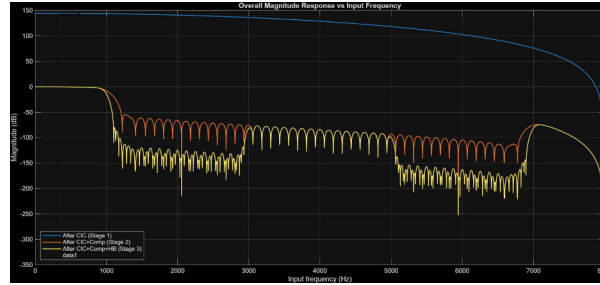


Figure 5: **Stage-wise Magnitude Response:** Blue: CIC only (Droop). Orange: CIC+Comp (Flat). Yellow: Final (Flat + Stopband).

5.3 Coefficient Quantization

FIR stages designed using the **Parks–McClellan** algorithm compensate CIC passband droop and achieve desired passband ripple and stopband attenuation. In the 3rd-order $\Delta\Sigma$ modulator, a 4-bit quantizer improves loop stability and enlarges the linear range.

$$\text{SNR} \propto (2^B)^2 \cdot \text{OSR}^{2L+1} \quad (1)$$

6 HDL Design

6.1 RTL Architecture and Block Descriptions

The RTL-based digital decimation chain consists of three major stages:

1. CIC Filter
2. Multiphase Compensation Filter
3. Multiphase Half-Band (MHB) Filter.

6.2 Overall Architecture

The input stream $X[n]$ enters the CIC filter operating at the highest clock domain (512 kHz). Successive down-samplers reduce the sampling rate by a factor of 2 across the chain:

- **CIC Filter:** 512 kHz
- **Compensation Filter:** 8 kHz
- **MHB Filter:** 256 kHz internal clock for FIR MAC operations

Each down-sampling stage ($\downarrow 2$) discards alternate samples, thereby halving the data rate. The final output $Y[n]$ is both decimated and spectrally corrected.

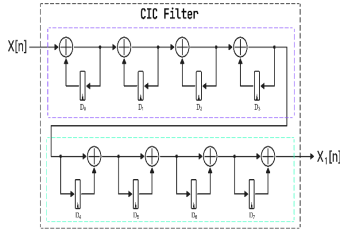


Figure 6: CIC Filter Architecture - 4th stage

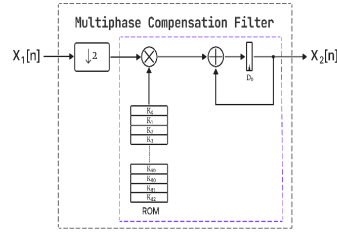


Figure 7: Multiphase FIR Compensation Filter Architecture

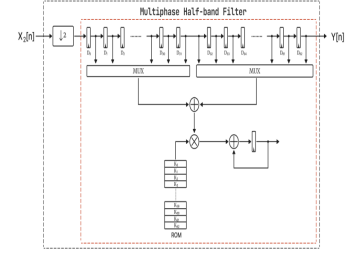


Figure 8: Multiphase Half-Band FIR Filter Architecture

6.3 Block Descriptions

1. CIC Filter

The CIC stage (Fig. 6) consists of four integrator stages running at 512 kHz, followed by a down-sampler and four comb stages operating at 8 kHz. **Purpose:** Provides large decimation efficiently, with internal growth determined by $(RM)^N$.

Integrator Section: Accumulates input samples using simple adders. Implements the recursion:

$$y[n] = y[n - 1] + x[n].$$

Comb Section: Applies backward differences after decimation:

$$y[n] = x[n] - x[n - RM].$$

2. Multiphase Compensation Filter

This block corrects the CIC passband droop.

- The down-sampler ($\downarrow 2$) brings the input rate to 8 kHz.
- A ROM stores coefficients K_0 to K_{62} .
- A multiplier, adder, and delay D_0 implement a low-complexity FIR structure.

3. Multiphase Half-Band (MHB) Filter

This is the final decimation and channel-shaping stage.

- A decimator halves the rate again, while the internal datapath runs at 256 kHz to complete all FIR MAC operations per output sample.
- A half-band FIR structure (with many zero-valued coefficients) uses delays D_0 to D_{62} .

6.4 Simulation Results and Observations

The complete testbench environment was used to verify the sigma-delta ADC and the decimation filter. The testbench internally generates the input stimulus, applies the sigma-delta modulator behavior, and processes the resulting bitstream through the CIC + HF filter. The waveform shown in the (Fig. 9) which has simulation result.

A sinusoidal input with added flicker noise was applied, and the resulting 20-bit decimated output values were observed. The waveform confirms the expected behavior across all stages:

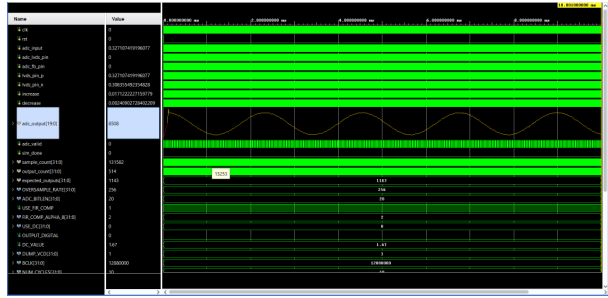


Figure 9: Output response of Digital Filter

6.5 Timing Analysis Results

The design was analyzed at a system clock frequency of 100 MHz (10 ns period) even though actual design will run at 512 KHz. The static timing results show **WNS = +0.524 ns** and **WHS = +0.116 ns**, indicating that both setup and hold timing requirements are comfortably met with no violations.

Timing	Setup	Hold
Worst Negative Slack (WNS):	0.524 ns	
Total Negative Slack (TNS):	0 ns	
Number of Failing Endpoints:	0	
Total Number of Endpoints:	4485	
Implemented Timing Report		

Timing	Setup	Hold
Worst Hold Slack (WHS):		0.116 ns
Total Hold Slack (THS):		0 ns
Number of Failing Endpoints:		0
Total Number of Endpoints:		4485
Implemented Timing Report		

(a) Worst Negative Slack Result

(b) Worst Hold Slack Result

Figure 10: Timing Analysis Results

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7 ASIC Synthesis & Layout Results

7.1 Synthesis Results (Area & Timing)

Design Metrics:

- **Design:** Filter_Top
- **Version:** X-2025.06
- **Library:** tsl18fs120_scl_ss
- **Number of References:** 28
- **Number of Ports:** 112
- **Number of Nets:** 810
- **Number of Cells:** 731 (577 comb., 151 seq., 0 macros)
- **Number of Buffer/Inverter:** 150
- **Final Line Count:** 1

Area Report and Timing Report

Hierarchical area and timing breakdown with optimized standard-cell mapping:

Table 6: ASIC Area Report: Hierarchical Breakdown

Item	Area (m²)
Combinational Area	10,473.890063
Buffer/Inverter Area	990.689996
Noncombinational Area	9,097.050255
Macro/Black Box Area	0.000000
Net Interconnect Area	545.190814
Total Cell Area	19,570.940318
Total Area	20,116.131132

(a) Area Report

Table 7: Static Timing Analysis: Critical Path and Slack

Timing Parameter	Incr (ns)	Path (ns)
Clock Domain: clk (Rise Edge)		
Clock Period	1000.00	1000.00
Clock Network Delay (ideal)	0.00	0.00
Clock Uncertainty	-26.00	980.00
Slack Analysis		
Library Setup Time	-0.16	979.84
Data Required Time	—	979.84
Data Arrival Time	—	113.35
Slack (MET)	—	866.48 ns

(b) Timing Report

Figure 11: Timing and Area Analysis Results

Table 6: Static Timing Analysis: Critical Path and Slack

Timing Parameter	Incr (ns)	Path (ns)
Clock Domain: clk (Rise Edge)		
Clock Period	1000.00	1000.00
Clock Network Delay (ideal)	0.00	0.00
Clock Uncertainty	-26.00	980.00
Slack Analysis		
Library Setup Time	-0.16	979.84
Data Required Time	—	979.84
Data Arrival Time	—	113.35
Slack (MET)	—	866.48 ns

7.2 Layout Summary (Utilization & Congestion)

Utilization Analysis

Core utilization achieved 72.22% with balanced distribution across M1–M3 metal layers. The remaining 27.78% capacity is reserved for clock tree synthesis (CTS), signal routing, and buffering, representing an optimal balance between cell density and routing resources.

Utilization Interpretation: A 72.22% core utilization ensures adequate routing resources without overcrowding. The remaining 27.78% provides sufficient space for:

Congestion Snapshot/Report

Final congestion map as mention in the (Fig. 12) after global routing shows the majority of regions fall into the lowest congestion bin (blue, count 1740), indicating minimal overflow and a low-congestion design with balanced distribution across M1–M3 metal layers. Overall overflow is less than 5%, confirming successful routing closure.

8 Challenges Faced

8.1 Technical and Design Challenges Encountered

- A $\Sigma\Delta$ ADC using a **CIFB modulator architecture** was selected based on stability, modulator order, quantizer resolution, and compatibility with **CIC–FIR decimation**. Most literature provides conceptual guidance, requiring additional effort for a stable, synthesizable design.

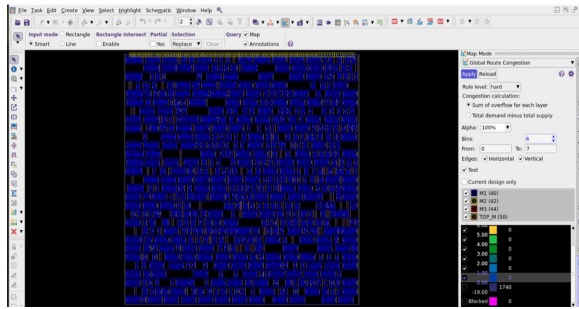


Figure 12: Global routing congestion map. Blue regions (count: 1740) indicate minimal overflow with balanced M1–M3 track usage.

```

*****
Report : congestion
Design : Filter_Top
Version: U-2022.12-SP3
Date   : Sun Dec  7 18:58:33 2025
*****

```

Layer	overflow		# GRCs has	
Name	total	max	overflow (%)	max overflow
Both Dirs	0	0	0 (0.00%)	0
H routing	0	0	0 (0.00%)	0
V routing	0	0	0 (0.00%)	0

Figure 13: Global routing congestion report.

- Design optimization involved trade-offs among **power, area, complexity, and robustness**, requiring iterative refinement to meet performance targets.
- Target **ENOB** was achieved by balancing **OSR, modulator order, and quantizer resolution**. Noise shaping reduced in-band noise but increased out-of-band noise, placing stringent requirements on the **CIC and compensation FIR filters**.
- The absence of unified resources covering **architecture selection, stability analysis, quantizer design, and decimation** required extensive cross-referencing of multiple sources.

9 New Learning

9.1 Key Learnings from Modelling, RTL, and ASIC Flow

PVT and Environment Setup Validated designs across all **PVT corners** using foundry **.lib** files and established an **NDM-based physical design environment** with complete technology and constraint integration.

Physical Design Flow Implemented the full **physical design flow**, including floorplanning, placement, CTS, routing, and **GDSII generation**, with focus on timing, area, power, and routability.

Filter Architecture Trade-offs Evaluated $\Sigma\Delta$ **modulator architectures and orders**, studying feedback-/feedforward paths and integrator scaling effects on **stability, noise shaping, and ENOB**.

Noise Analysis and Mitigation Analyzed major **noise sources** and applied mitigation techniques such as **noise shaping**, supply isolation, decoupling, and robust clocking.

10 Limitations & Risks

10.1 Noise and Stability Concerns

This section analyzes delta–sigma modulator architectures capable of achieving 16–19 bit ENOB at Nyquist rates of 0.5–2 ksp/s. We evaluate discrete-time, continuous-time, and hybrid topologies using MATLAB/Simulink behavioral models to study performance trade-offs, noise behavior, and sensitivity to non-idealities such as thermal noise, flicker noise, and clock jitter.

10.2 Fixed-Point Precision Errors

- Limited hardware required fixed-point (e.g., Q-format) coefficients, introducing precision loss that affects filter accuracy, linearity, and SNR.
- These errors appear as quantization noise, coefficient distortion, and round-off effects; thus, we used larger word lengths, guard bits, proper scaling, and rounding to minimize them.

11 References

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